CNENGR202 Instrumentation and Control

Part 2 – L4 Strain gauges, Piezo, Induction

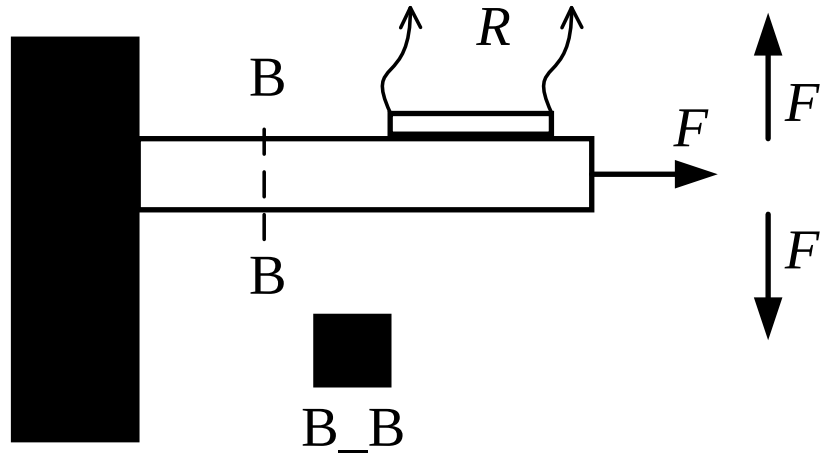
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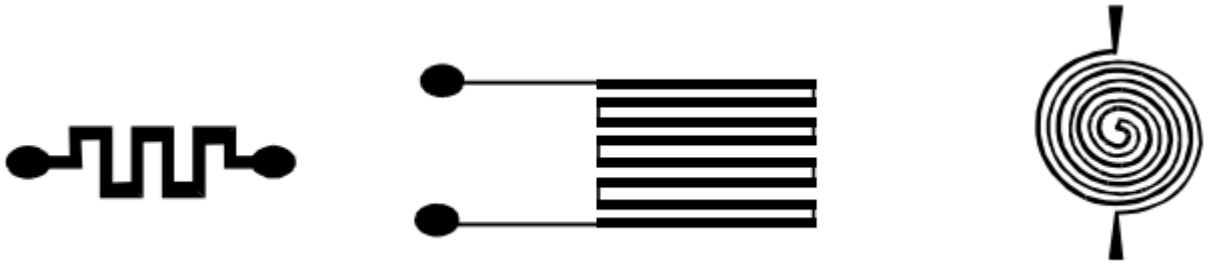
School of Electronics and Information Engineering
Beijing Jiaotong University

Resistance Strain Gause

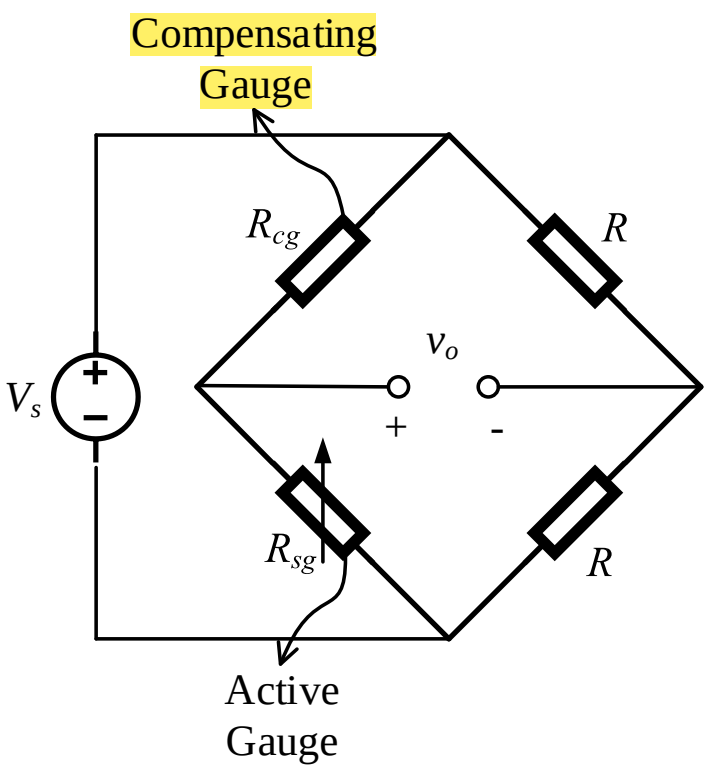
$$R = \rho \frac{l}{S}$$

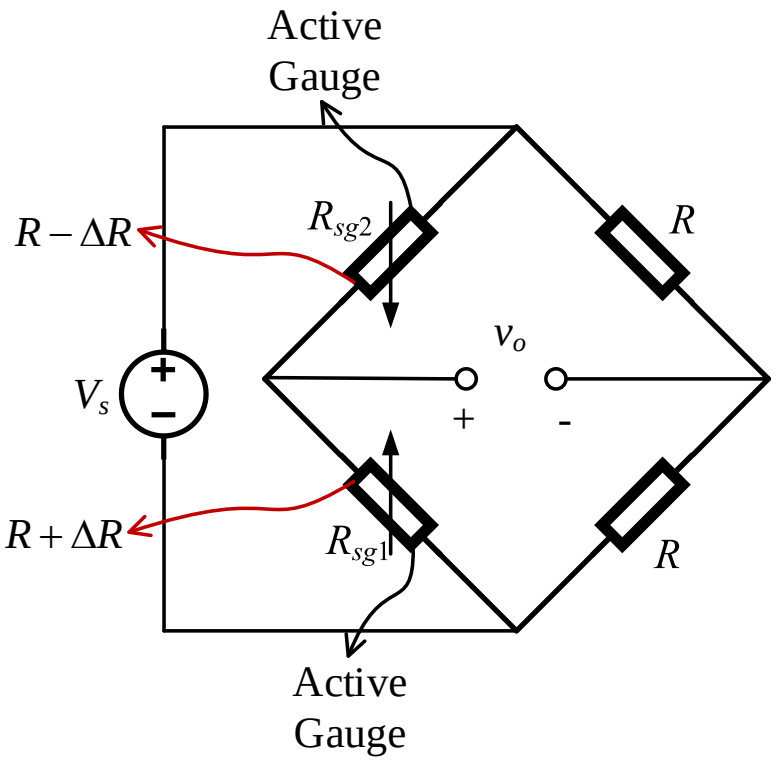


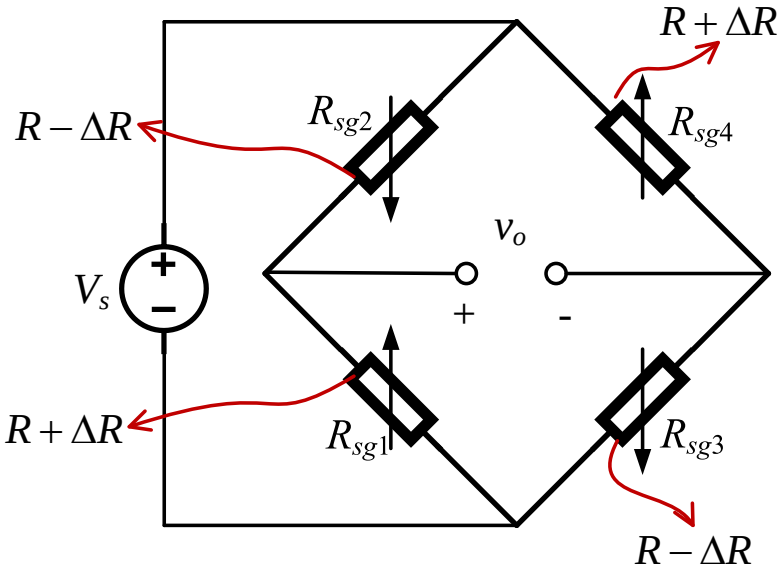
Metallic resistance gauges



Measuring circuit

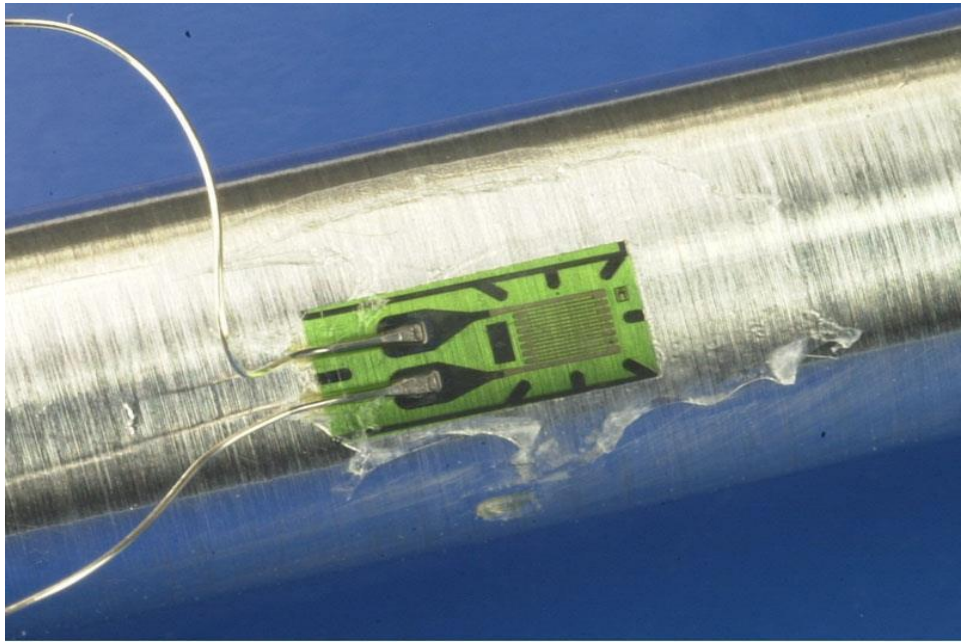






Typical strain gauge properties

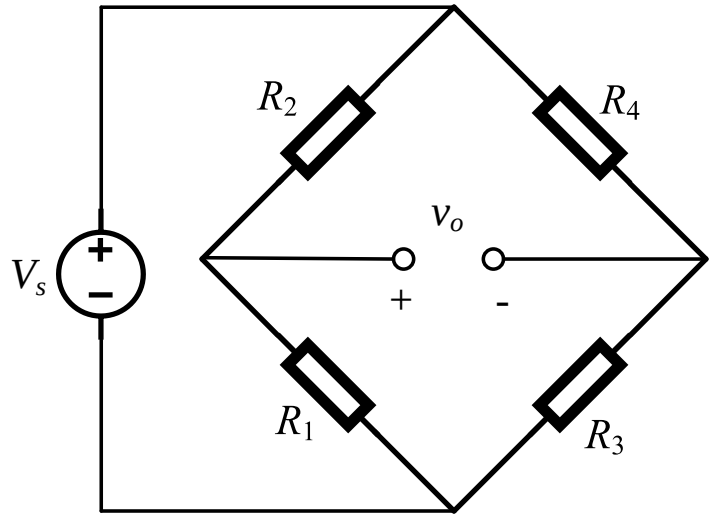
- Gauge factor: 2
- Unstrained resistance: 120Ω
- Linearity: +/- 0.3%
- Maximum tensile strain: +0.02
- Max compressive strain: -0.01
- Max operating temperature: 150^0C
- Total range: +4.8Ω to -2.4Ω



Class Questions 1

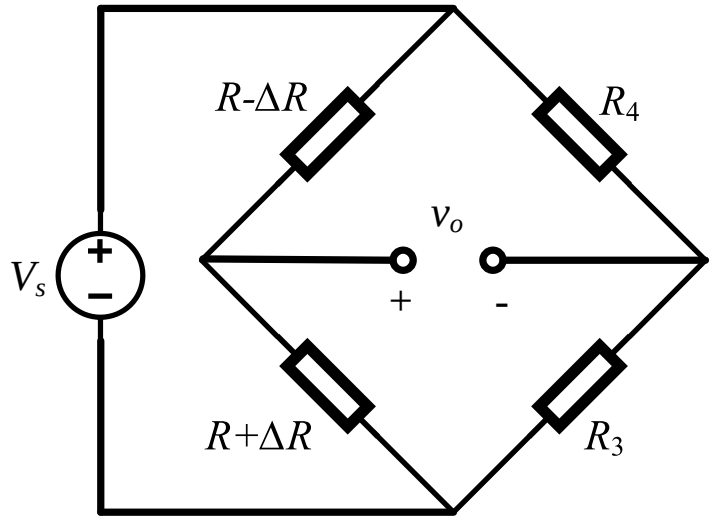
- If both R_1 and R_2 are replaced by strain gauges; one measuring the compression and one measuring the tension on the other side.

- What does the voltage equal now?



Answer for Class Questions 1

$$\frac{V_o}{V_s} = \frac{R+\Delta R}{R+\Delta R+R-\Delta R} - \frac{R}{R+R} = \frac{\Delta R}{2R}$$



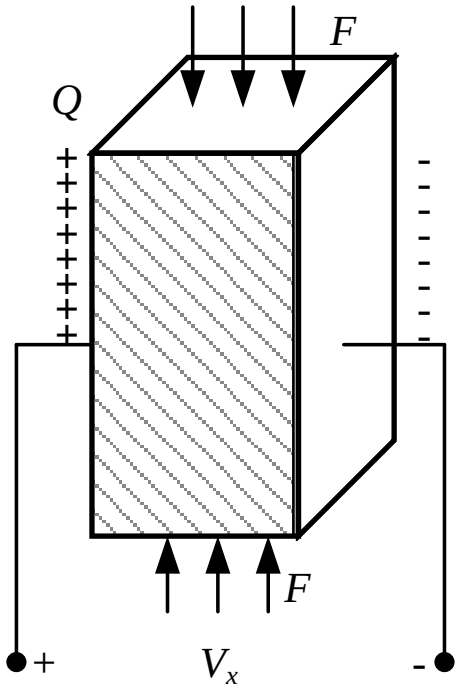
Piezoelectric Sensor

- When a electric charge Q is established in the dielectric crystal
- When a mechanical force is applied to the crystal

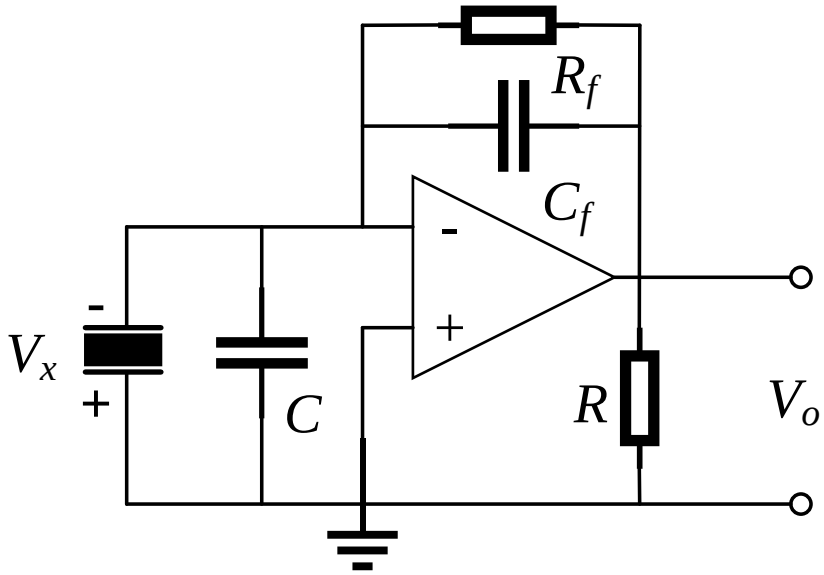
} Piezoelectric effect

$$V_x = \frac{Q}{C_x}$$

- V_x : Electric voltage
- C_x : Electrical capacity of the sensor with metallic plates



Measuring circuit



$V_o = ?$

Home Question

- Attaching one end of a heavy rope to the wall and a strain gauge half way along the rope (and assume it sticks). This strain gauge goes into a Wheatstone bridge circuit and then onto a voltmeter for reading. If the following information is available, how much force are you all producing between you?

- Nominal resistance of strain gauge: 100Ω
- Resistance of other resisters in Wheatstone bridge: 100Ω
- Voltage into Wheatstone bridge: 1V
- Maximum Voltage out of Wheatstone bridge: 1mV

